

SURAJ BALAJI RAUTRAO

AI Engineer | MLOps Engineer | Data Scientist

+49 17665689275 • surajrautao02@gmail.com • linkedin.com/in/surajrautao • github.com/SurajRautao • Magdeburg, Germany | Work Permit: Germany

WORK EXPERIENCE

Eaton Corporation | Data Engineer

Oct 2022 – Aug 2024

- Contributed to global automotive market intelligence initiatives, driving analytics and process optimisation across business units.
- Built a Power BI dashboard to detect systematic downward revisions in long-term BEV production forecasts, enabling executive leadership to recalibrate strategic growth assumptions.
- Architected a centralised Power BI platform integrating complex multi-source APAC vehicle production data, replacing fragmented regional reporting and improving strategic decision-making.
- Engineered a Python-based ETL automation integrating data from 20+ SharePoint sources into a unified strategic plan template, improving operational efficiency by 75%.

Flextronics Technologies Pvt. Ltd. | Engineer

May 2018 – Sep 2022

- Ensured 100% accurate and timely loading of Master Data into Oracle Agile PLM, supporting cross-functional product lifecycle management processes.
- Developed a Power BI dashboard for inventory classification, enabling supply chain managers to optimise warehouse evaluation and allocate resources to high-value, high-variability items.

EDUCATION

MSc in Digital Engineering

2024 – Present

Otto-von-Guericke-Universität Magdeburg, Germany

Bachelor of Engineering in Mechanical Engineering

2017

Savitribai Phule Pune University, India

PROJECTS

ScholarScope – Multi-Agent AI Research Assistant — Python · LangGraph · RAG · Ollama · FastAPI · Streamlit · Docker

- Built a multi-agent AI research assistant using LangGraph and Retrieval-Augmented Generation (RAG) for automated literature review and citation analysis.
- Integrated local LLMs with research paper retrieval APIs to generate structured summaries and interactive citation graphs.
- Developed a full-stack architecture using Streamlit, FastAPI, Docker, NetworkX, and PyVis.

Customer Churn Prediction & Explainable AI Dashboard — CatBoost · FastAPI · Airflow · MLflow · SHAP · Docker · Streamlit

- Designed and deployed an MLOps pipeline for customer churn prediction using CatBoost and FastAPI, with automated training workflows in Apache Airflow.
- Created an interactive Streamlit dashboard with SHAP-based explainability for local and global feature interpretation; tracked experiments with MLflow.
- Containerised the application using Docker and deployed cloud-hosted API and dashboard services.

Document Classification & Explainable AI using LIME and Local LLMs — TF-IDF · Doc2Vec · SBERT · SVM · ANN · LIME | Group Project, OVGU Magdeburg

- Evaluated TF-IDF, Doc2Vec, and SBERT embeddings across SVM, ANN, and Decision Tree classifiers on 20 Newsgroups and AG News datasets.
- Applied LIME for model explainability and benchmarked results against LLM-generated reasoning.

Fake Review Detection using LLMs and RAG — LangChain · FAISS · MiniLM · Llama 3 · Gemma 3 · Phi 4 · Ollama | Digital Engineering, OVGU Magdeburg

- Built an NLP-based fake review detection system using LLMs, Prompt Engineering, and RAG achieving up to 96% accuracy on benchmark datasets.
- Evaluated Llama 3, Gemma 3, and Phi 4 using zero-shot, one-shot, and few-shot prompting strategies; developed a LangChain + FAISS retrieval pipeline with MiniLM embeddings.

- Automated evaluation workflows using Python, Ollama, and GPU-based execution environments.

SKILLS

Machine Learning: Supervised & Unsupervised Learning, Neural Networks, Explainable AI (LIME/SHAP), Retrieval-Augmented Generation (RAG), Prompt Engineering, Zero/One/Few-Shot Learning

MLOps: MLflow, Apache Airflow, FastAPI, LangGraph, LangChain, Streamlit, Docker, Ollama

LLMs & NLP: Llama 3, Gemma 3, Phi 4, SBERT, Doc2Vec, TF-IDF, FAISS, MiniLM, CatBoost

Python: Pandas, NumPy, Scikit-learn

Power BI: DAX, Power Query, Data Modeling, Dataflows, Scheduling and Security Management

SQL: Joins, CTEs, Subqueries, Window Functions, Stored Procedures, Views

LANGUAGES

German: B1

English: C1